

RESULTS

Results

01 · PROPENSITY SCORE MATCHING

treatment effect via matching

Table 1. Covariate balance (before / after)

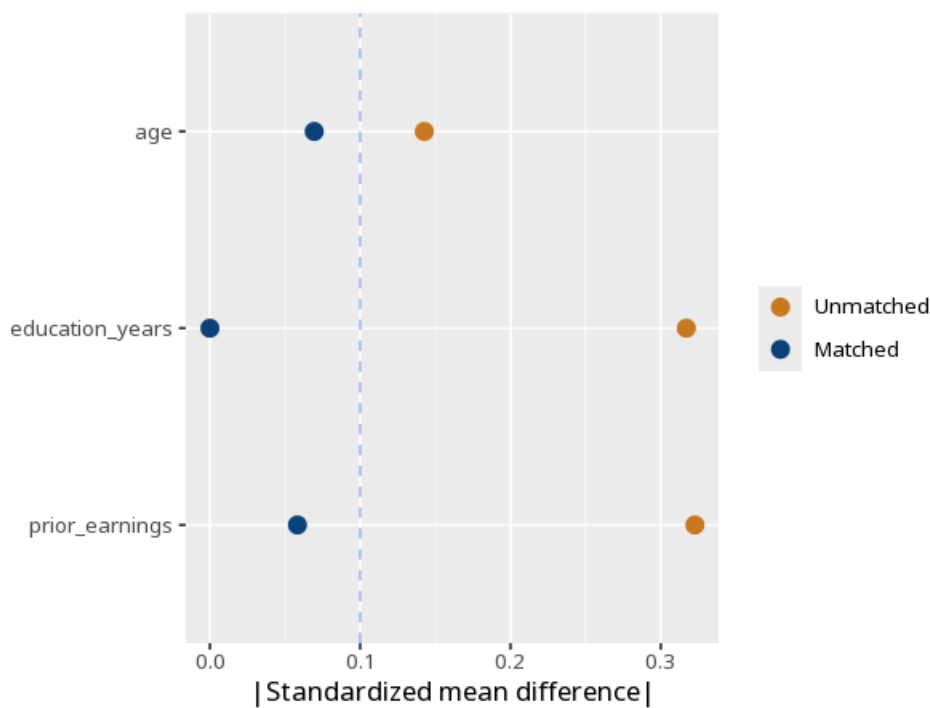
Covariate	Std. mean diff (pre)	Std. mean diff (post)	Variance ratio
age	−0.14	0.07	1.03
education_years	−0.32	0.00	0.97
prior_earnings	−0.32	−0.06	1.04

Table 2. Treatment effect (ATT)

	(1)
Treatment (ATT)	1948.76 (549.28) [856.07, 3041.46]
Num.Obs.	84
Treated (matched)	42
Control (matched)	42
Common support: 0 treated unit(s) dropped (off common support)	
Propensity overlap: treated [0.06, 0.30], control [0.05, 0.30]	

good matching drives post-matching standardized mean differences toward 0 (commonly < 0.1). The common-support rows report how many treated units fell off the region of common support (dropped) and the propensity-score overlap range by group — wide non-overlap weakens the comparison.

Figure. Balance



Understanding the numbers

Covariate. Which covariate this balance row reports on. Here, this row is for age.

Std. mean diff (pre). The standardized mean difference on this covariate before matching. Here, the pre-matching SMD is -0.14 .

Std. mean diff (post). The standardized mean difference after matching - good matching drives this toward 0 (commonly below 0.1). Here, the post-matching SMD is 0.07.

Variance ratio. The ratio of treated-to-control variance on this covariate after matching - closer to 1 means more similar spread. Here, the variance ratio is 1.03.

ATT. The average treatment effect for the treated units, estimated after matching. Here, the ATT is 1948.76.

Num.Obs.. The total number of matched observations the ATT is based on. Here, $N \text{ matched} = 84$.

Treated (matched). The number of treated units retained after matching. Here, $\text{treated (matched)} = 42$.

Control (matched). The number of control units retained after matching. Here, $\text{control (matched)} = 42$.

How to read this test

First confirm balance — matched groups should look alike on the covariates (small standardized differences). Then read the ATT as the treatment effect for the treated, with p/CI. The common-support rows tell you how many treated units were dropped for lacking a comparable control and where the two groups' propensity scores overlap — good overlap with few drops means the ATT generalizes to most of the treated. PSM only balances the covariates you measured and included, so the causal reading also assumes no important confounder was left unmeasured (ignorability) — an

assumption that cannot be verified from the data. Method: R MatchIt package (Ho, Imai, King & Stuart, 2011); balance diagnostics (Austin, 2011). Your significance threshold (α) is 0.05.

APA template: After propensity-score matching, the ATT was 1948.76, 95% CI [856.07, 3041.46], $p < .001$.

Statistical basis: Propensity score matching (Ho DE, Imai K, King G, Stuart EA (2011). "MatchIt: Nonparametric Preprocessing for Parametric Causal Inference." Journal of Statistical Software, 42(8), 1-28.); Covariate-balance diagnostics (Austin, P. C. (2011). "An Introduction to Propensity Score Methods for Reducing the Effects of Confounding in Observational Studies." Multivariate Behavioral Research, 46(3), 399-424.). Full references in the exported CITATIONS.txt.

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